A Recurrent-Exponent Self-Similarity Criterion for the $\ell^\infty(L^{p_k})$ Factorization Problem

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Status

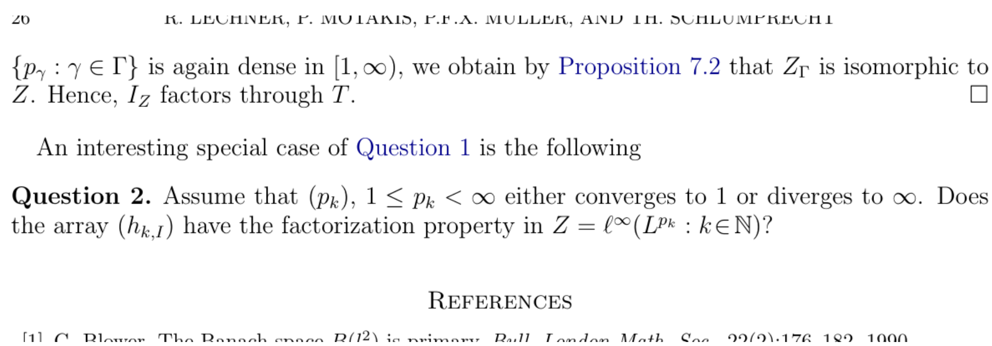
This is a partial result for the final questions of Lechner, Motakis, Mueller, and Schlumprecht [1]. It proves a broad self-similar mixed-exponent case of the large-diagonal factorization problem for the Haar array in

$$Z(p) = \ell^\infty(L^{p_k} : k \in \mathbb{N}).$$

It does not settle the strict endpoint cases $p_k \to 1$ or $p_k \to \infty$ from Question 2 of the source paper.

Source Question

The source paper asks the following on page 26.



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$\{p_\gamma : \gamma \in \Gamma\}$ is again dense in $[1,\infty)$, we obtain by Proposition 7.2 that Z_Γ is isomorphic to Z. Hence, I_Z factors through T. □

An interesting special case of Question 1 is the following

Question 2. Assume that (p_k), $1 \le p_k < \infty$ either converges to 1 or diverges to ∞. Does the array $(h_{k,I})$ have the factorization property in $Z = \ell^\infty(L^{p_k} : k \in \mathbb{N})$?

References

[1] C. Blower. The Banach space $B(l^2)$ is primary. *Bull. London Math. Soc.*, 22(2):176–182, 1990.

In text form:

> Assume that (p_k), $1 \le p_k < \infty$, either converges to 1 or diverges to ∞. Does the array $(h_{k,I})$ have the factorization property in $Z = \ell^\infty(L^{p_k} : k \in \mathbb{N})$?

The present packet proves a nearby structural criterion that recovers the dense-exponent example in the source paper and extends it to arbitrary recurrent exponent lists.

Definitions

For a sequence $p = (p_k)$ in $[1,\infty)$, write

$$Z(p) = \ell^\infty(L^{p_k} : k \in \mathbb{N}).$$

The Haar array in $Z(p)$ is denoted by $(h_{k,I})_{k\in\mathbb{N}, I\in\mathcal{D}}$, where $h_{k,I}$ is the usual L^∞-normalized Haar function h_I placed in the k-th coordinate.

Definition 1. *An exponent sequence $p = (p_k) \subset [1, \infty)$ is* recurrent *if, for every $k \in \mathbb{N}$ and every $\varepsilon > 0$, the set*

$$\{m \in \mathbb{N} : |p_m - p_k| < \varepsilon\}$$

is infinite.

Proof Intuition

The source paper proves a powerful subsequence theorem: under the Haar hypotheses, every operator on $Z(p)$ with large diagonal factors the identity on a sub-sum $Z(p|_\Gamma)$, and Γ can be forced to meet any prescribed countable family of infinite sets. To turn this into a factorization of $I_{Z(p)}$, one needs $Z(p|_\Gamma)$ to be another copy of $Z(p)$.

Recurrence is exactly a clean way to guarantee this. It lets us prescribe infinitely many future coordinates whose exponents approximate each original exponent p_k to the finite-dimensional accuracy needed in Proposition 7.2 of the source paper. The Johnson-ultrafilter construction used there then embeds each L^{p_k} complementably into the selected sub-sum. Running this coordinatewise and applying Pelczynski decomposition gives $Z(p|_\Gamma) \cong Z(p)$. Composing this isomorphism with the source factorization on $Z(p|_\Gamma)$ gives the desired factorization of $I_{Z(p)}$.

Main Result

Theorem 1. *Let $p = (p_k) \subset [1, \infty)$ be recurrent. Then the Haar array $(h_{k,I})_{k,I}$ has the factorization property in $Z(p) = \ell^\infty(L^{p_k} : k \in \mathbb{N})$. In other words, if $T : Z(p) \to Z(p)$ is bounded and*

$$\inf_{k,I} |h^*_{k,I}(Th_{k,I})| > 0,$$

then $I_{Z(p)}$ factors through T.

Lemma 1. *Let $p = (p_k)$ and $q = (q_k)$ be sequences in $[1, \infty)$. Assume:*

1. *for every k, p_k is an accumulation value of the sequence q;*
2. *for every k, q_k is an accumulation value of the sequence p;*
3. *$Z(p)$ is isomorphic to $\ell^\infty(Z(p))$.*

Then $Z(q)$ is isomorphic to $Z(p)$.

Proof. We first recall the argument of Proposition 7.2 in [1]. If a fixed exponent r is an accumulation value of q, choose pairwise distinct indices m_s such that $q_{m_s} \to r$ and so fast that the identity map between the first s Haar levels of $L^{q_{m_s}}$ and L^r is a 2-isomorphism. The map

$$f \in L^r \longmapsto (R_s f)_{s=1}^\infty \in \ell^\infty(L^{q_{m_s}})$$

where R_s is the s-th Haar partial-sum projection, is then an isomorphic embedding. The ultrafilter coefficient-limit map used in Proposition 7.2 gives a bounded left inverse, so the image is complemented. The constants are uniform after fixing the numerical tolerance 2.

Applying this construction for each coordinate p_k, and using disjoint subsequences of the q-coordinates, gives a complemented embedding of $Z(p)$ into $Z(q)$. The same argument with the roles of p and q interchanged gives a complemented embedding of $Z(q)$ into $Z(p)$. Since $Z(p) \cong \ell^\infty(Z(p))$, Pelczynski decomposition yields $Z(q) \cong Z(p)$. □

Lemma 2. *If p is recurrent, then $Z(p) \cong \ell^\infty(Z(p))$.*

Proof. Let p^∞ denote the sequence obtained by repeating every p_k countably many times. Then $Z(p^\infty)$ is isometric to $\ell^\infty(Z(p))$. The sequence p^∞ trivially has each p_k as an accumulation value, and recurrence of p lets us allocate disjoint subsequences of p approximating each countably repeated exponent in p^∞. Thus the proof of Lemma 1, applied to p and p^∞, gives complemented embeddings both ways. Since $Z(p^\infty)$ is visibly isomorphic to its own ℓ^∞-sum, Pelczynski decomposition gives $Z(p) \cong Z(p^\infty) \cong \ell^\infty(Z(p))$. □

Proof of Theorem 1. For each pair (k,s), choose $\eta_{k,s} > 0$ so small that whenever $|r - p_k| < \eta_{k,s}$, the coefficient identity between the first s Haar levels of L^r and L^{p_k} is a 2-isomorphism. Put

$$\Omega_{k,s} = \{m \in \mathbb{N} : |p_m - p_k| < \eta_{k,s}\}.$$

By recurrence, every $\Omega_{k,s}$ is infinite.

Theorem 2.5 of [1] applies to the Haar array in $Z(p)$: the Haar bases have uniformly bounded basis constants, the required diagonal factorization property, and simultaneous strategic reproducibility, as recorded in the source paper for the L^p Haar systems. Therefore, for any operator $T : Z(p) \to Z(p)$ with large diagonal, there is an infinite $\Gamma \subset \mathbb{N}$ such that $\Gamma \cap \Omega_{k,s}$ is infinite for every k, s, and

$$I_{Z(p|_\Gamma)}$$

factors through T.

The intersection property implies that every exponent p_k is an accumulation value of the subsequence $p|_\Gamma$. Conversely, every exponent in $p|_\Gamma$ is one of the original p_k's, hence is an accumulation value of p by recurrence. By Lemmas 1 and 2,

$$Z(p|_\Gamma) \cong Z(p).$$

Composing this isomorphism with the source factorization of $I_{Z(p|_\Gamma)}$ through T yields a factorization of $I_{Z(p)}$ through T. This proves the theorem. □

Limitations

The theorem does not resolve Question 2 as stated. A strict sequence $p_k \to 1$, or a strict sequence $p_k \to \infty$, usually has many isolated exponents: for a fixed early p_k, there may be no infinite tail of exponents approximating p_k. The argument above then cannot prove that the selected sub-sum $Z(p|_\Gamma)$ is isomorphic to the original $Z(p)$. Equivalently, the remaining endpoint obstruction is an absorption problem for isolated L^p-coordinates inside tails whose exponents converge to 1 or to ∞.

Novelty and Literature Check

The source paper itself proves the dense-exponent case by showing that if (p_k) is dense in $[1,\infty)$, then the selected sub-sum remains isomorphic to $Z(p)$. The present packet abstracts that mechanism to a recurrence/self-similarity criterion. Local checks of later factorization-related sources in the run cache, including arXiv:2011.09915, 2102.10088, 2112.12534, and 2605.21711, found related factorization, primariness, and Haar-system results but no direct answer to the endpoint question and no exact statement of this recurrent-exponent criterion. Web searches for phrases around arXiv:1910.11188, $\ell^\infty(L^{p_k})$, Haar arrays, and the endpoint Question 2 also found no direct answer beyond the source paper itself.

Human Review Recommendation

Review as a likely valid partial result. The key points to check are:

1. the adaptation of Proposition 7.2 of [1] from dense exponent sequences to arbitrary cluster sequences;

2. the coordinatewise complemented-embedding construction using disjoint approximating subsequences;

3. the Pelczynski decomposition step converting mutual complemented embeddings into an isomorphism.

References

[1] R. Lechner, P. Motakis, P. F. X. Mueller, and Th. Schlumprecht, *The factorization property of* $\ell^\infty(X_k)$, arXiv:1910.11188, 2019.